Shikang Wen

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Hong Kong/Berkeley

RESEARCH INTERESTS

- Carbon neutral building
- Deep learning
- AI-driven building design

EDUCATION EXPERIENCE

Bachelor of Architecture, Shanghai Jiao Tong University 2018.09 - 2023.06
PhD candidate, The Hong Kong Polytechnic University 2023.09 - Present
Visiting Scholar, University of California, Berkeley 2026.05 - Present

ACADEMIC PUBLICATIONS

1. **S. Wen**, X. Hu, G. Hua, P. Xue, and D. Lai, Comparing the performance of four shading strategies based on a multi-objective genetic algorithm: A case study in a university library, 2022. (**Journal of Building Engineering, IF: 8.1**)
2. **S. Wen**, J. Xue, G. Hua, and D. Lai, Reconstructing fisheye luminance maps with a two-step network from a single low dynamic range image, 2024. (**Automation in construction**, **IF: 12.6**)
3. **S. Wen**, R. You, and Q. Chen, Implementing building retrofitting strategies to halve campus office building carbon emissions by 2035: A case study in Hong Kong with techno-economic analysis, 2025. (**Energy and Buildings**, **IF: 8.0**)
4. **S. Wen**, B. Shen, Y. Yao, and Q. Chen, Energy performance and life-cycle analyses of advanced terminal devices and heat pumps for a medium-sized office building in different climate zones in China, 2025. (**Energy and Buildings**, **IF: 8.0**)
5. **S. Wen**, R. You, and Q. Chen, Energy-exergy performance and indoor comfort of a radiant ceiling cooling system with air-to-air sensible heat exchanger in hot and humid climate, 2025. (**Building and Environment**, **IF: 8.4**)
6. **S. Wen** and Q. Chen, A novel building design method through 2D-to-3D reconstruction: Reconstructing from 2D images to 3D building energy modeling inputs, 2026. (**Advanced Engineering Informatics, IF: 11.5**)
7. **S. Wen**, R. You, and Q. Chen, An AI-driven design for energy-efficient buildings: A case study in a campus office building, 2026. (**Applied Energy, Under Review**)

RESEARCH EXPERIENCE

1. Optimization of Building daylit space Mar 2021 - Jun 2022

Under the guidance of Prof. Lai, I developed a framework to compare various shading strategies, incorporating a performance curve generated from a multi-objective genetic algorithm. This experience sharpened my research, simulation, optimization, and academic writing skills.

Contribution: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.

2. Real-time glare detection Jul 2022 - Mar 2023

An academic project, supervised by Prof. Lai and Prof. Xue, I developed a deep learning approach to reconstruct luminance maps from single-exposure images, honing my skills in generative adversarial networks (GANs) and high dynamic range (HDR) photo analysis.

Contribution: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.

3. Building retrofits in high-density campus Sep 2023 - May 2024

Under the supervision of Prof. Chen and Prof. You during my PhD project, I developed expertise in building retrofitting strategies, research methodology, building energy simulation tools, and academic writing.

Contribution: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.

4. Advanced radiant ceiling cooling system May 2024 - Jan 2025

Under the supervision of Prof. Chen during my PhD project, I mastered the analysis of radiant cooling systems and specialized in problem-solving research methodologies.

Contribution: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.

5. AI-driven energy-efficient building design Feb 2025 - Jun 2025

During my doctoral studies under the supervision of Prof. Chen and Prof. You, I mastered AI-driven building design and specialized in developing automated building generation to the EnergyPlus model.

Contribution: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.

6. AI-based 2D image to 3D EnergyPlus model May 2025 - Nov 2025

During my doctoral studies under Prof. Chen's supervision, I specialized in image-to-3D models and developed automated systems for generating building forms—transforming single 2D images into EnergyPlus geometry inputs.

Contribution: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.

7. AI-based scalable personal comfort HVAC control system. Dec 2025 - Sep 2026

During my doctoral studies under the supervision of Prof. Chen, I specialized in computational fluid dynamics (CFD) simulations and developed personal comfort HVAC control systems for scalable applications in building environments.

Contribution: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.

HONORS & AWARDS

PolyU International Collaborative Research Fellowship	2026/07
IAQVEC 2026 Student Fellowship	2026/05
National Inspirational Scholarship	2022/12
Shanghai Jiao Tong University Scholarship	2022/12
National Inspirational Scholarship	2021/12
Shanghai Jiao Tong University Scholarship	2021/12
Third Prize in Shanghai Jiao Tong University Structure Competition	2020/12
Second Prize in the 2020 Li Zhengdao Science and Art Competition	2020/11
Third Prize in Shanghai Jiao Tong University Structure Competition	2019/11
First Prize in the 2019 Li Zhengdao Science and Art Competition	2019/11
Second Prize in Tongji University Wooden Structure Competition	2019/05

PROFESSIONAL SERVICE

- **External reviewer** in Journal of Building Engineering (since 2022)
- **External reviewer** in Building and Environment (Since 2025)

Talk

- Invited to present on AI-driven building design at the WeLL Seminar, Center for the Built Environment (CBE), UC Berkeley. - Oct 2026
- Invited to give a talk at Arizona State University (ASU) on AI in building energy modeling. - Sep 2026
- Presented on radiant cooling systems in hot and humid climates at the BEEE Research Salon, PolyU. - Feb 2026

PROGRAM LANGUAGES OR SOFTWARE SKILL

- Python
- Grasshopper
- EnergyPlus
- CFD

LANGUAGES

- English
- Chinese